Autopoietic Cohomology: Iterative Obstruction Repair, Manifold Emergence, and Gauge Structure on Causal Posets

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Abstract

Autopoietic cohomology describes a process in which unresolved ℓ^1 coboundary defects induce the minimal augmentation of the underlying causal structure required for their resolution.

We define a discrete dynamical system — the **Autopoietic Iterator** — that executes this structural extension by iteratively repairing cohomological obstructions on a causal poset. At each step, the system detects the leading obstruction class $[o_n] \in H^n(\mathcal{P}; F)$, computes a minimal-cost transport extension Λ_n via ℓ^1 optimization, and forms the augmented poset $\mathcal{P}_{n+1} = \mathcal{P}_n \coprod_\partial \Lambda_n$. Because defect resolution is achieved through structural extension rather than internal reconfiguration, the construction is strictly monotone (the poset only grows) and the coboundary norm is non-increasing. We observe that the repair LP admits a formulation closely related to the Kantorovich optimal transport problem, making each repair step a discrete W transport optimization.

This construction synthesizes the obstruction results of [1], the ℓ^1 classification theorems of [2, 3], the cohomological structure of [4], and the universal framework of [5]. We propose — as a suggestive analogy — that this iterative repair process provides a candidate dynamical mechanism for Deutsch's Constructor Theory [6], in which physical evolution corresponds to sequential resolution of topological inconsistencies.

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1. Introduction

1.1 Background

In the preceding papers:

- **Paper 000** [1] and **Paper 003** [4] established that projection dynamics generate coboundary defects, persisting with canonical cohomological structure.
- **Paper 001** [2] and **Paper 002** [3] classified admissible diagnostics on Banach presheaves and finite graphs, showing ℓ^1 total variation is uniquely selected.
- **Paper 004** [5] synthesized these into the Universal Obstruction Theory framework, extending to operator-valued settings and W duality.

The missing piece is **dynamics**: given a poset with non-trivial obstructions, what process resolves them, and what structure does it generate?

1.2 The Autopoietic Structural Principle

We formalize the principle that **defect resolution is not internal optimization—it is structural extension**. If an ℓ^1 defect cannot be reduced within an existing poset, geometric consistency demands the augmentation of the underlying morphism structure.

We define the **Autopoietic Iterator** as the formal discrete dynamical system mapping this process: it iteratively detects unresolvable obstruction classes and augments the causal poset to resolve them via minimal ℓ^1 transport geometry.

This paper defines the iterator precisely and proves its basic structural properties.

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2. Mathematical Construction

2.1 The Autopoietic Iterator

Let $\mathcal{P}_0$ be a finite causal poset with presheaf $F_0 : \mathcal{P}_0^{\mathrm{op}} \to \mathbf{Ban}$, and suppose $I(s) = \sum_e \|\delta_e(s)\|_1 > 0$ (non-trivial coboundary norm).

Definition 2.1 (Autopoietic Iterator). The sequence $\{\mathcal{P}_n\}_{n\geq 0}$ is defined recursively:

(i) Detect. Compute the leading obstruction class $[o_n] \in H^{k_n}(\mathcal{P}_n; F_n)$, defined as the class with the maximal ℓ^1 norm contribution to the global stable norm.

(ii) Optimize. Compute the optimal repair morphism:

$$\Lambda_n = \arg\min_{\Lambda} \; I_{\mathcal{P}_n \cup \Lambda}(s)$$

where the minimization is evaluated over all admissible candidate additions $\Lambda \subset (V \times V) \setminus E_{\mathrm{cov}}(\mathcal{P}_n)$ satisfying boundary compatibility conditions restricted to $\partial(\mathrm{supp}([o_n]))$, and I is the ℓ^1 coboundary norm.

(iii) Augment. Form the pushout:

$$\mathcal{P}_{n+1} = \mathcal{P}_n \coprod_{\partial(\mathrm{supp}([o_n]))} \Lambda_n$$

where the pushout is taken along boundary inclusions in the category of finite posets with monotone maps.

(iv) Terminate. If $I_{\mathcal{P}_{n+1}}(s) = 0$ (all obstructions resolved), halt. Otherwise, return to (i).

Remark. The optimization step (ii) is an ℓ^1 linear program. For finite posets, it is solvable via standard linear programming methods (polynomial-time in the size of the discretized problem). The pushout in step (iii) produces a strictly larger poset.

2.2 The Complexity Ratchet

Definition 2.2 (Monotone Growth Constraint). The Autopoietic Iterator satisfies the **Complexity Ratchet**: the inclusion $\mathcal{P}_n \hookrightarrow \mathcal{P}_{n+1}$ is strict, and all existing covering relations are preserved. No morphisms may be deleted:

$$E_{\mathrm{cov}}(\mathcal{P}_n) \subseteq E_{\mathrm{cov}}(\mathcal{P}_{n+1}), \qquad |\mathcal{P}_{n+1}| > |\mathcal{P}_n| \text{ whenever } I_{\mathcal{P}_n}(s) > 0$$

Remark 2.2 (Modeling assumption). The monotone growth constraint is a design choice, not a mathematical necessity. We restrict attention to monotone augmentations to model discrete irreversibility. However, it is critical to note that **monotone growth does not imply monotone simplification of cohomology**. Introducing new causal edges to repair a target obstruction can inadvertently introduce secondary obstructions across the expanded topology.

2.3 Basic Properties

Proposition 2.1 (Monotone Cost Reduction). *If Λ_n is chosen to minimize $I_{\mathcal{P}_n \cup \Lambda}(s)$, then:*

$$I_{\mathcal{P}_{n+1}}(s) \leq I_{\mathcal{P}_n}(s)$$

with equality only if no repair morphism reduces the coboundary norm. If additionally Λ_n trivializes the leading obstruction class (i.e., $[o_{k_n}]|_{\mathcal{P}_{n+1}} = 0$), the corresponding stable-norm summand vanishes from the cost.

Proof. The identity morphism $\Lambda = \emptyset$ (adding no new edges) is always a feasible choice, yielding $I_{\mathcal{P}_n \cup \emptyset}(s) = I_{\mathcal{P}_n}(s)$. Since Λ_n is chosen as the minimizer over all admissible Λ, we directly obtain $I_{\mathcal{P}_{n+1}}(s) \leq I_{\mathcal{P}_n}(s)$.

For the stronger claim: if $[o_{k_n}]|_{\mathcal{P}_{n+1}} = 0$, then by the cost formula (Theorem 2 of [4]), the k_n-th summand $\|[o_{k_n}]\|_1$ vanishes in the augmented poset. The remaining summands $\|[o_k]\|_1$ for $k \neq k_n$ may change, but the total cost remains bounded by the minimality property. $\square$

Open Questions.

1. **Convergence.** Does the iterator terminate in finitely many steps for all finite posets, or can it cycle indefinitely? Monotone norm decrease does not guarantee termination, since the ℓ^1 cost takes values in a continuous range and could decrease asymptotically. For bounded-degree posets with integer-valued stalks, finite termination is guaranteed, but the general case is open.

2. **Uniqueness.** Is the optimal Λ_n unique, or does a highly symmetric $\mathcal{P}_n$ accommodate multiple cost-equivalent minimal extensions?
3. **Complexity.** What is the formal computational complexity bound of the full discrete trace acting as a function of $|\mathcal{P}_0|$?

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3. Algorithmic Interpretation

This framework dictates a dual purpose: first, as a theoretical foundation, and second, as a concrete, executable computing structure. The abstract iterator definition provides a directly implementable algorithmic protocol.

3.1 Discrete Wasserstein Descent

The repair optimization in step (ii) of the iterator assumes the structural configuration of the **Kantorovich optimal transport problem** [9]. The stable norm computation:

$$\|[o_n]\|_1 = \min_{\omega} \sum_{e} |\omega(e)| \quad \text{subject to} \quad \omega = o_n + \delta f$$

is the Kantorovich dual of optimal transport: minimizing the ℓ^1 cost to move defect mass subject to a continuous divergence (coboundary) constraint.

This framework	Optimal transport
Defect cocycle o_n	Mass imbalance $\mu - \nu$
Cochain correction δf	Transport plan γ
ℓ^1 coboundary norm	Wasserstein-1 cost
LP dual (max-flow)	Kantorovich dual

This identification arises naturally from the dual structure of the ℓ^1 optimization. A discrete W transport problem operates iteratively by moving topological defects toward minimal boundaries.

Connection to minimum cut. Since the ℓ^1 coboundary norm scales proportionally to the graph total variation, each repair optimization maps back cleanly to a minimum cut extraction on the defect support graph. Inserting the optimal Λ_n matches inserting a structural bridge across the min-cut boundary separating conflicting defect regions. Each descent step evaluates computationally within polynomial time via established algorithms like push-relabel.

Why the ℓ^1 Geometry. The ℓ^1 norm naturally yields sparse, localized, LP-solvable repairs within this framework. An evaluated ℓ^2 integration diffuses defects globally throughout the entire connected lattice structure. The ℓ^1 formulation provides targeted minimum cuts and transport mechanics, facilitating a constrained, executable descent.

3.2 The Omega Engine as Evidence

An illustrative computational implementation of the iterative formalism reinforces this. Preliminary observations (not formal benchmarks) suggest the **Omega Engine** (`anomalon_kernel`) instantiates the Autopoietic Iterator in silico:

- **Detect:** Locate ℓ^1-maximized cohomological defects iteratively across the active graph state.
- **Optimize:** Execute the associated ℓ^1 linear objective to chart the sparsest minimum transport routing.
- **Augment:** Compile the proposed modification into structural code boundaries mimicking the posetal pushout.

If the iterator strictly operates as a transport optimization, the engine should exhibit specific, computationally verifiable signatures: 1. **Defect transport:** Cohomological obstruction density will map deterministically along minimal cost constraints evaluated iteratively. 2. **Min-cut concentration:** Codebase or topological graph adaptations should concentrate overwhelmingly across min-cut margins rather than randomized surface areas. 3. **Sparse repair:** Active topological integrations consistently display finite, targeted revisions in isolated regions.

These measurable programmatic behaviors provide structural evidence consistent with the model's formulation.

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4. Connection to the Paper Series

The Autopoietic Iterator functionally unifies the preceding anomaly architecture:

Paper	Result used
[1] / [4] — Obstruction & Cohomology	Obstructions exist and persist; $I(s) > 0$ forms the active initial condition.
[2] / [3] — ℓ^1 Classification	The cost functional I is uniquely determined for boundary isolation.
[5] — Universal Framework	Wasserstein-1 framework identifies repair optimization as explicit transport tracking.

> *The Autopoietic Iterator evaluates functionally as a discrete dynamical system minimizing the persistent ℓ^1 coboundary norm by integrating cost-optimal topological modifications.*

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5. Physical Conjectures

The following interpretations are entirely speculative. They are proposed as suggestive analogies, motivated exclusively by the structural topology of the framework, and do not present mathematically established mappings to conventional physical systems.

5.1 Time as Ordinal Iteration

Conjecture 1 (Speculative). *Physical time correlates dimensionally with the ordinal progression count n of the discrete Autopoietic Iterator. The irreversibility modeled by discrete thermodynamic entropy translates topologically via the Complexity Ratchet constraint.*

This remains a primarily philosophical conjecture. Evaluating it against standard formulations (like Lorentz invariance and relativistic causality structures) would demand deriving continuous limit sequences and demonstrating macroscopic time symmetries currently absent from the construction.

5.2 Geometry from Obstruction Density

Conjecture 2 (Speculative). *Macroscopic geometry evaluates fundamentally from the concentration density determining continuous obstruction integrations. Deep recursion within a specific dense obstruction boundary implies severe geometric curvature anomalies—a topological analogue to gravity.*

Providing this hypothesis formalized rigor necessitates modeling emergent phenomena mirroring Einstein field configurations, deriving explicit limits from localized Kan extension concentrations.

5.3 Decoherence as Truncation

Conjecture 3 (Speculative). *This serves as a suggestive analogy, not a derived model of quantum measurement: Wavefunction collapse heuristically maps to the forced truncation of the ∞-groupoid occurring wherever localized processing saturation mathematically exceeds internal structural capacity boundaries dictated by local Lieb-Robinson bounds. Exceeding dimension thresholds prompts sudden truncation down to a singular definable 1-topos representation (a discrete outcome).*

This represents the broadest metaphysical speculation included herein. Canonical decoherence dictates strict environmental trace mechanics and explicit reductions of entangled density boundaries. Correlating this mathematically requires defining analogous environment structures within the poset and proving the theoretical reduction accurately reproduces Born probability thresholds.

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6. Discussion

This paper formally branches from Papers 1–4 by transitioning the established topological metrics into an active dynamic processing definition.

The mathematical formulation provides: - **Definition 2.1:** Defines the active iterator via sequential [1] defect targeting and posetal pushouts. - **Definition 2.2:** Codifies irreversible evolution explicitly via the Complexity Ratchet assumption. - **Proposition 2.1:** Confirms global cumulative optimization guarantees assuming monotonic edge construction.

Additionally, isolating the Algorithmic Interpretation highlights that this formalism constructs explicit programmatic pipelines (evidenced by the AutoP anomaly compiler logic), bridging abstract topologies with testable discrete behaviors like localized minimum cuts.

Evaluating the physical conjectures depends strictly on solving defining bounds around iterator termination constraints, extracting computable continuous analogues, and executing wide-scale computational profiling of localized engine integrations. It strictly presents a mapped hypothesis outlining discrete defect-driven thermodynamic and chronological behaviors.

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7. References

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